Polynomial Mixing for Efficient Self-Supervised Speech Encoders

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Introduction

Context and motivation

- Self-supervised speech encoders usually rely on Transformer [11] or Conformer [5] architectures, and therefore inherit the **quadratic time and memory cost of multi-head attention (MHA) with respect to sequence length**.
- Can be particularly restrictive for long utterances, large-batch training, and efficient deployment [13].
- *Can we replace attention by a linear-complexity mixer while preserving ASR performance ?*

Contributions

- Introducing the **Polynomial Mixer** (PoM) as a replacement for the self-attention mechanism for speech encoders.
- Demonstrating the effectiveness of PoM as a replacement for MHA inside **a Conformer network trained using BEST-RQ** [2].
- Benchmarking PoM against standard MHA variants and linear-complexity baselines.
- Speech encoders equipped with PoM achieve a **competitive trade-off between WER, run-time and memory** on *LibriSpeech-100h* after fine-tuning.

Background

Baselines – Efficient sequence mixers

- **MHA** [11] models all pairwise token interactions explicitly.
- Popular MHA variants such as relative positional attention (TransformerXL) [3] and RoPE [10] still have quadratic complexity in the sequence length.
- **Reducing complexity by approximating or avoiding pairwise attention**, e.g.
  - Sparse attention : Longformer [1], BigBird [15]
  - MLP-based mixers : HyperMixer [7]
  - Global token mixers with sequence averaging : FastFormer [14], SummaryMixing [9]
  - State-space models : Mamba [4].

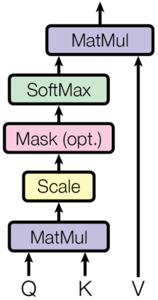
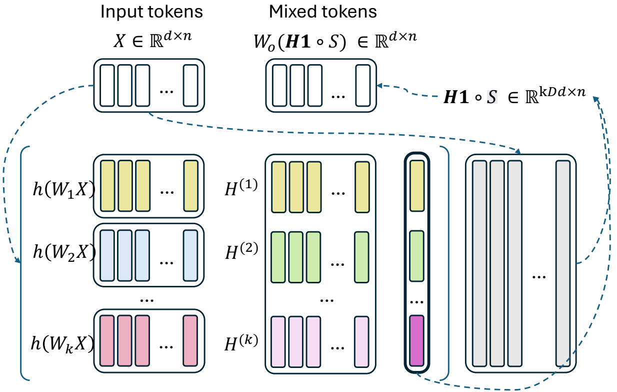


Figure 1 – In MHA, computing the scaled dot-product attention has quadratic cost in time and memory because of the $QK^{\top}$ product. Figure adapted from [11].

Method

Introducing the Polynomial Mixer (PoM) for Speech



$H = [H^{(1)} | \dots H^{(k)}] \in \mathbb{R}^{kDd \times n}$

$H^{(i)} = h(W_1X) \circ h(W_2X) \circ \dots \circ h(W_iX) \in \mathbb{R}^{Dd \times n}$

Sum: $\boldsymbol{H1} \in \mathbb{R}^{kDd \times 1}$

$S = \sigma(W_SX) \in \mathbb{R}^{kDd \times n}$

- Input seq. X : n tokens of dim. d
- Hyperparameters : polynomial order k, expansion factor D
- learnable weights $W_1, \cdots W_k \in \mathbb{R}^{Dd \times d}$
- $h(\cdot)$: GELU in our implem.
- $H(X)$: global state derived from the whole sequence
- selector projection matrix $W_s \in \mathbb{R}^{kDd \times d}$
- output projection matrix $W_o \in \mathbb{R}^{d \times kDd}$

Introducing PoM

Given an input sequence $X \in \mathbb{R}^{d\times n}$, PoM defines **a polynomial feature map** up to **degree** k in an expanded space of **factor** D :

$$H(X) = \Big[h(W_1X) \mid h(W_1X) \circ h(W_2X) \mid \dots \mid \prod_{m=1}^{k} h(W_mX)\Big]\mathbf{1}.$$

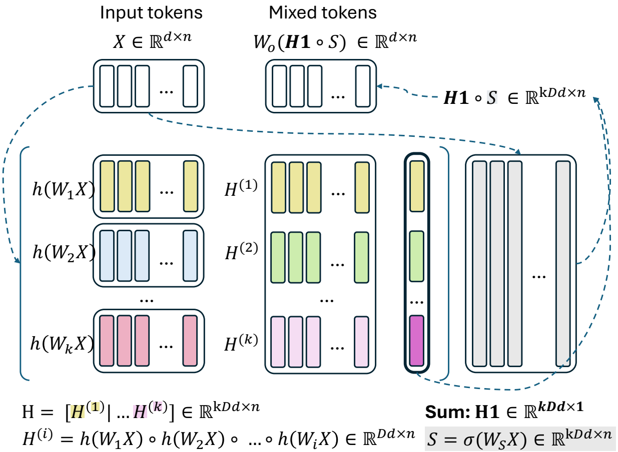
Averaging over time results in a **a single, sequence-level representation**, that is then broadcast over another projection W_SX of the input sequence (= token-wise selection). Finally, the sequence is mapped back to its original dimension.

$$\mathrm{PoM}(X) = W_o\Big[\sigma(W_sX) \circ H(X)\mathbf{1}^\top\Big].$$

Residual connections and feed-forward layers are retained. Replacement block for attention :

$$P(X) = X + \mathrm{PoM}(X) + \mathrm{FF}\big(X + \mathrm{PoM}(X)\big).$$

Complexity



Complexity

- **PoM** : shared state and per-token gating give $\mathcal{O}(n)$ scaling in sequence length.
- Each degree adds higher-order Hadamard interactions between projected token features.
- Every token sees the *same* shared state $\rightarrow$ token-specificity happens through the gate $\sigma(W_sX)$.

Results

Experimental protocol

Data and training (SpeechBrain [a])

- Pre-training : LibriSpeech 960h [8], BEST-RQ.
- Fine-tuning : LibriSpeech *train-100h*, ASR with CTC loss and an MLP decoder.
- Evaluation : *test-clean*, *test-other*
- Test time decoding : with and without language model decoding (kenLM [6])

Metric

Word Error Rate (WER)

Model sizes

- "Base" about 95M parameters
- "Large" about 315M parameters

Compared mixers

- PoM and its variants
- Quadratic : MHA vanilla, RoPE, relative-position MHA
- Linear alternatives : SummaryMixing, Mamba†, HyperConformer†, FastFormer†, following the comparative study of Whetten et al. [13].

a. https://github.com/speechbrain/speechbrain

BEST-RQ self-supervised learning (SSL) framework

BEST-RQ

- Input : Mel filter-bank frames
- Pre-training : pretext task = predicting codebook targets assigned to masked frames.
- Encoder backbone : Conformer, which alternates
  - feed-forward module,
  - **token mixer** (normally attention),
  - convolution module.

What changes here ?

- We keep the BEST-RQ framework and Conformer architecture
- Only the *attention module* is replaced.

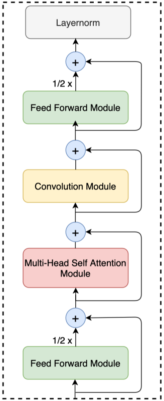


Figure 2 – Architecture of a Conformer block. Figure adapted from [5].

Main ASR results : $\sim$ 95M models

- Among linear mixers, PoM is competitive, especially on *other* and with LM decoding.
- It approaches the MHA family, but does not fully match the best attention variants.

	Model	Clean	Clean + LM	Other	Other + LM
~95M	RelPosMHA [3]	**7.96 (± 0.32)**	**4.89 (± 0.25)**	**17.61** (± 0.54)	12.13 (± 0.44)
	RoPE MHA [10]	**8.06** (± 0.31)	**4.90** (± 0.26)	**17.53 (± 0.48)**	**11.98 (± 0.45)**
	vanilla MHA [11]	8.59 (± 0.32)	5.37 (± 0.25)	19.44 (± 0.54)	13.47 (± 0.46)
	PoM base	8.31 (± 0.31)	<u>5.42</u> (± 0.27)	<u>19.06</u> (± 0.53)	<u>13.62</u> (± 0.48)
	SummaryMixing [9]	9.79 (± 0.34)	5.93 (± 0.27)	22.80 (± 0.60)	15.84 (± 0.51)
	Mamba† [4]	<u>7.61</u>	<u>5.50</u> (± 0.28)	19.97	15.37
	HyperConformer† [7]	8.22	5.77 (± 0.28)	19.29	15.03
	FastFormer† [14]	9.32	6.82 (± 0.31)	22.75	17.95

Table 1 – “Base” models (95M parameters). WER on LibriSpeech *test-clean* and *test-other* after BEST-RQ pre-training on LibriSpeech-960h and fine-tuning on *train-100*. Lower is better. **Best MHA variant** in bold, <u>best linear mixer</u> underlined.

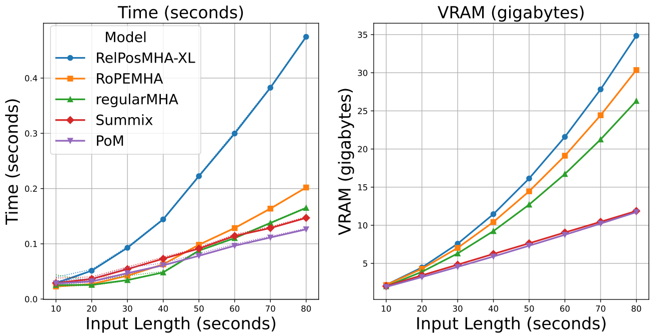
Main ASR results : $\sim$ 315M models

- Scaling up improves all models, but the gap between PoM and the best MHA variants remains visible.
- PoM is competitive, although not SotA in terms of WER for large models.

	Model	Clean	Clean + LM	Other	Other + LM
$\sim$315M	RelPosMHA [3]	**4.92 ($\pm$ 0.25)**	**3.49 ($\pm$ 0.21)**	**10.78 ($\pm$ 0.37)**	**8.09 ($\pm$ 0.35)**
	RoPE MHA [10]	5.13 ($\pm$ 0.26)	3.66 ($\pm$ 0.21)	10.99 ($\pm$0.33)	8.45 ($\pm$ 0.37)
	PoM base	6.28 ($\pm$ 0.26)	<u>4.52</u> ($\pm$ 0.23)	14.86 ($\pm$ 0.47)	<u>11.33</u> ($\pm$ 0.43)
	SummaryMixing [9]	7.35 ($\pm$ 0.31)	4.85 ($\pm$ 0.25)	17.60 ($\pm$ 0.53)	12.97 ($\pm$ 0.49)
	Mamba† [4]	<u>5.59</u>	<u>4.48</u> ($\pm$ 0.25)	15.47	12.66
	HyperConformer† [7]	5.87	<u>4.54</u> ($\pm$ 0.32)	<u>13.13</u>	<u>10.78</u>
	FastFormer† [14]	13.16	9.89 ($\pm$ 0.34)	31.91	26.75

Table 2 – "Large" models (315M parameters). WER on LibriSpeech *test-clean* and *test-other* after BEST-RQ pre-training on LibriSpeech-960h and fine-tuning on *train-100*. Lower is better. **Best MHA variant** in bold, <u>best linear mixer</u> underlined.

Efficiency of PoM



- From 10 s to 80 s, MHA runtime and memory increase much faster.
- At 80 s, PoM uses about 2.8× less memory than relative-position MHA.
- Runtime is close to SummaryMixing and faster than RoPE MHA in this setup.

Discussion

Interpretation

- PoM replaces self-attention by a *global polynomial state* + *token-wise selector*. No pairwise attention matrix is formed.
- Speech encoders may not always need full pairwise token interaction (NB : not new).
- PoM's global state / sequence summary vector already captures useful structure for ASR, and encode interactions between projected features.
- Polynomial interactions are richer than a single arithmetic mean.

Limitations

- PoM cannot form token-specific pairwise alignments.
- The same state is broadcast to every token.
- Remaining gap compared to the best attention variants, especially at larger scale.
- Limited scope (English, ASR)

Conclusion

Recap – Results

- PoM is an **efficiency-oriented alternative to MHA** and proposes a new trade-off between **expressivity and efficiency** for SSL speech representation learning.
- PoM has **linear complexity** in sequence length and integrates naturally into Conformer-style speech encoders.
- On LibriSpeech-100h, it outperforms or performs close to other linear mixers in terms of WER.
- The best MHA variants still achieve lower WER, especially in the larger model regime, but at a significantly higer cost in terms of runtime and memory.
- Attention is of course not obsolete for speech SSL, but *a simpler linear mixer can get reasonably close while saving memory and time* .

Perspectives

- Hybrid encoders that would use attention in early layers and PoM in upper layers
- Experimenting on other languages than English
- Experimenting on other tasks e.g. speaker recognition, emotion recognition

Questions ?

References i

I. Beltagy, M. E. Peters, and A. Cohan.
Longformer : The long-document transformer.
arXiv preprint arXiv:2004.05150, 2020.

C.-C. Chiu, J. Qin, Y. Zhang, J. Yu, and Y. Wu.
Self-supervised learning with random-projection quantizer for speech recognition.
In *International Conference on Machine Learning*, pages 3915–3924. PMLR, 2022.

Z. Dai, Z. Yang, Y. Yang, J. G. Carbonell, Q. Le, and R. Salakhutdinov.
Transformer-xl : Attentive language models beyond a fixed-length context.
In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2978–2988, 2019.

A. Gu and T. Dao.
Mamba : Linear-time sequence modeling with selective state spaces.
arXiv preprint arXiv:2312.00752, 2023.

A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu, et al.
Conformer : Convolution-augmented transformer for speech recognition.
2020.

K. Heafield, I. Pouzyrevsky, J. H. Clark, and P. Koehn.
Scalable modified kneser-ney language model estimation.
In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 2 : Short Papers)*, pages 690–696, 2013.

F. Mai, A. Pannatier, F. Fehr, H. Chen, F. Marelli, F. Fleuret, and J. Henderson.
Hypermixer : An mlp-based low cost alternative to transformers.
In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1 : Long Papers)*, pages 15632–15654, 2023.

- V. Panayotov, G. Chen, D. Povey, and S. Khudanpur.
  Librispeech : an asr corpus based on public domain audio books.
  In *2015 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pages 5206–5210. IEEE, 2015.
- T. Parcollet, R. van Dalen, S. Zhang, and S. Bhattacharya.
  Summarymixing : A linear-complexity alternative to self-attention for speech recognition and understanding.
  In *Proc. Interspeech 2024*, pages 3460–3464, 2024.
- J. Su, M. Ahmed, Y. Lu, S. Pan, W. Bo, and Y. Liu.
  Roformer : Enhanced transformer with rotary position embedding.
  Neurocomputing, 568:127063, 2024.

A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin.
Attention is all you need.
Advances in neural information processing systems, 30, 2017.

R. Whetten, T. Parcollet, M. Dinarelli, and Y. Estève.
Open implementation and study of best-rq for speech processing.
In *2024 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW)*, pages 460–464. IEEE, 2024.

R. Whetten, T. Parcollet, A. Moumen, M. Dinarelli, and Y. Estève.
An analysis of linear complexity attention substitutes with best-rq.
In *2024 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2024.

C. Wu, F. Wu, T. Qi, Y. Huang, and X. Xie.
Fastformer : Additive attention can be all you need.
arXiv preprint arXiv:2108.09084, 2021.

M. Zaheer, G. Guruganesh, K. A. Dubey, J. Ainslie, C. Alberti, S. Ontanon, P. Pham, A. Ravula, Q. Wang, L. Yang, et al.
Big bird : Transformers for longer sequences.
Advances in neural information processing systems, 33:17283–17297, 2020.

BEST-RQ self-supervised learning (SSL) framework

Why BEST-RQ ?

- strong and efficient SSL method,
- open-source implementation in SpeechBrain [12]
- compatible with different token mixers

Principle

- Input : Mel filter-bank frames
- Pre-training : pretext task = predicting codebook targets assigned to masked frames.
- Encoder backbone : Conformer.

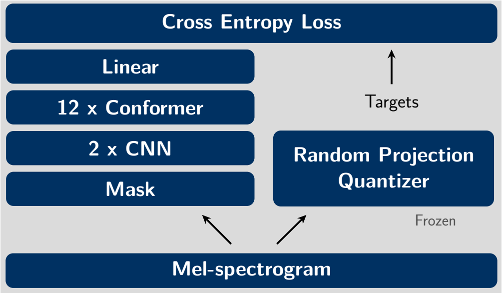


Figure 3 – Diagram of BEST-RQ architecture as proposed by Chiu et al. [2] and implemented by Whetten et al. [12]

Variants of PoM

Hypotheses for designing PoM

- Higher degree and expansion factor may increase expressivity.
- Separating frequency paths may be beneficial e.g. separating phonemic vs. semantic information.

PoM variants

1. **Base** : concatenate degrees 1 through k.
2. **Mode jump** : keep only the highest degree k.
3. **2-way / 3-way frequency splits** : mix groups of features separately, motivated by different spectral roles.

Ablation study

PoM variants

- The *highest-degree only* “jump” variant is slightly worse than the base version : **intermediary degrees are useful too** $\rightarrow$ The chosen “base” version preserves all moments by concatenating degrees $1, \ldots, k$.
- Selective mixing (e.g. separating high/low-freq. features) might help in underfitting settings / smaller networks, does not improve over full mixing (base version) $\rightarrow$ sparse selection or feature splitting does not significantly improve ASR performance.

PoM	clean	clean+LM	other	other+LM
base	8.76 ± 0.28	**5.51 ±0.15**	**19.87 ±0.51**	**14.02 ±0.32**
jump	8.74 ± 0.20	5.73 ± 0.18	20.56 ± 0.60	14.89 ± 0.68
2ways	8.75 ± 0.26	**5.65 ±0.11**	<u>20.28 ±0.57</u>	14.60 ± 0.31
3ways	8.77 ± 0.47	5.82 ± 0.11	20.86 ± 0.76	15.08 ± 0.42

Table 3 – WER on LibriSpeech test splits, averaged by PoM version for different model sizes.

Ablation study

Preliminary experiments

- Performance improves with larger effective capacity ($k \times D \times d$).
- Main performance factor (more than the PoM variant) : choice of k and D to define expressivity of the multi-degree polynomial summary
- Gains appear to saturate around degree $k = 3$ and expansion $D = 2$.
- Using 5% layer drop helps both MHA and PoM.

mixer	ldrop	clean	clean+LM	other	other+LM
PoM (all)	✗	8.92	5.71 (−3.21)	19.86	14.12 (−5.74)
PoM (all)	✓	8.61 (−0.31)	5.42 (−3.50)	19.81 (−0.05)	14.00 (−5.86)
MHA	✗	8.39	5.30 (−3.09)	19.01	13.25 (−5.76)
MHA	✓	8.20 (−0.19)	5.05 (−3.34)	18.19 (−0.82)	12.53 (−6.48)

Table 4 – WER averaged by mixer type, depending on the use of layerdrop during training (ldrop) and of a language model at inference (LM).

Scaled dot-product attention

Scaled Dot-Product (SDP) Attention

$X \in \mathbb{R}^{n \times d}$ input matrix

$W^Q \in \mathbb{R}^{d \times d_k}, W^K \in \mathbb{R}^{d \times d_k}, W^V \in \mathbb{R}^{d \times d_v}$

$Q = XW^Q \in \mathbb{R}^{n \times d_k}$ query matrix

$K = XW^K \in \mathbb{R}^{n \times d_k}$ key matrix

$V = XW^V \in \mathbb{R}^{n \times d_v}$ value matrix

$$\underbrace{Attention(Q, K, V)}_{\in \mathbb{R}^{n \times d}} = softmax\left(\frac{QK^{\mathsf{T}}}{\sqrt{d_k}}\right)V$$

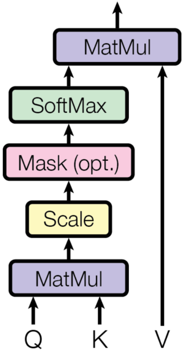


Figure 4 – Figure adapted from [11]

Multi-head attention

Multi-Head Attention (MHA)

Compute h attention “heads” in parallel:

$$\boldsymbol{head_i} = \boldsymbol{Attention}(\boldsymbol{QW_i^Q}, \boldsymbol{KW_i^K}, \boldsymbol{VW_i^V}) = \underbrace{softmax\left(\frac{QW_i^Q (KW_i^K)^\top}{\sqrt{d_k}}\right)}_{S_i} V_i$$

With $Q, K, V \in \mathbb{R}^{n \times d_m}$, $W_i^Q \in \mathbb{R}^{d_m \times d_k}$, $W_i^K \in \mathbb{R}^{d_m \times d_k}$, $W_i^V \in \mathbb{R}^{d_m \times d_v}$

Concatenate heads: $[head_1 | head_2 | .. | head_h] \in \mathbb{R}^{n \times h d_v}$

And multiply by $W^O \in \mathbb{R}^{h d_v \times d_m}$ to obtain the final output:

$$\boldsymbol{MHA}(\boldsymbol{X}) = [\boldsymbol{head_1} | \boldsymbol{head_2} | .. | \boldsymbol{head_h}] \boldsymbol{W^O} \in \mathbb{R}^{n \times d_m}$$

NB: usually, $d_k = d_v = d_m / h$ and Q, K, V are set to X in the above equations.

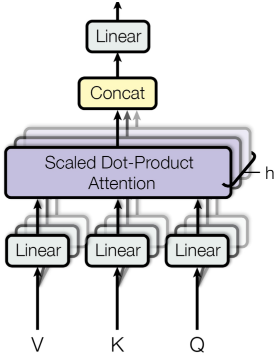


Figure 5 – Figure adapted from [11]

SummaryMixing by Parcollet et al. [9]

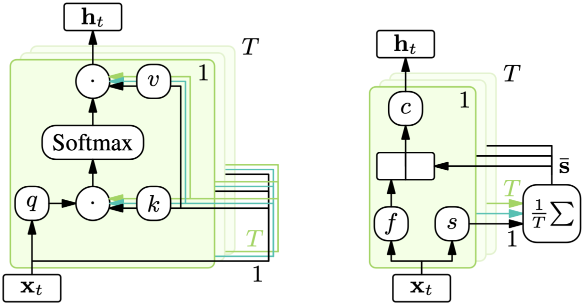


Figure 1: *Comparison of the self-attention cell (left) and the newly proposed SummaryMixing cell (right). In SummaryMixing, the information from all time steps is averaged, and this average is fed back to each time step T.*

Figure 6 – Figure from the authors [9]